

Delay-Robust Tube Model-Predictive Control for Six-DOF Spacecraft Rendezvous and Docking

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A delay-aware Tube Model Predictive Control (Tube-MPC) architecture is developed, bespoke to the six-degree-of-freedom (6-DOF) dynamics of a chaser spacecraft executing mid-field rendezvous and docking (RVD) with a target vehicle. Tube-MPC is motivated for delay-laden actuator channels, and spacecraft-specific linearization and coupling assumptions are presented. A Gramian-based weight-selection strategy is derived, invariant-set via a Lyapunov ellipsoid is established, and the approach is validated numerically. Results demonstrate that the controlled state trajectory converges to the docking equilibrium while the deviation remains inside an invariant tube despite time-varying actuator delays.

1 Introduction

Rendezvous and docking (RVD) between spacecraft demand precise translational and rotational control in the presence of actuator limits, sensing noise, and communication latency. Delays in the actuator or command channels produce a mismatch between computed control commands and actual applied thrusts, degrading tracking performance and endangering constraint satisfaction. Historically, remedies have included predictor-based compensators, Smith predictors for constant delays, and adaptive or robust MPC variants. However, these methods either use precise delay models or introduce excessive complexity for marginal robustness improvement.

Tube-MPC provides an alternative in which an *undisturbed nominal* trajectory is computed using a standard MPC, and a state-feedback policy is enforced around it to confine the actual trajectory to a bounded tube. This decomposition yields invariance and feasibility guarantees by leveraging a fixed nominal linear model for tube construction while allowing on-line re-linearization for prediction accuracy. Tube-MPC is particularly suitable for spacecraft RVD with bounded, variable actuator delays, offering quantifiable trade-offs between robustness, nominal precision, and computational cost.

The key contributions of this study are:

- A delay-aware Tube-MPC architecture explicitly tailored to coupled 6-DOF spacecraft dynamics with translational–attitude coupling.
- A Gramian-informed weight-normalization framework ensuring balanced modal controllability across heterogeneous state groups.
- A Lyapunov-based ellipsoidal invariant certificate validated numerically, demonstrating bounded robustness under variable actuator delays.

2 Problem statement and methodology

A chaser spacecraft operating at mid-field separation from its target is considered, executing an approach and docking manoeuvre within the local-vertical local-horizontal (LVLH) frame. The control objectives are:

- to stabilize the relative x -position at 10 m and other positions and attitudes to the docking equilibrium, $x_k \rightarrow 0$ which is considered the operating point (x_0, u_0) ;
- to respect actuator force/torque limits and state constraints;

- to guarantee robust invariance to delays modelled as an additive disturbance.

The adopted methodology is summarized as follows:

1. The nonlinear 6-DOF plant is linearized about the operating point (x_0, u_0) to obtain the nominal pair (A, B) .
2. An LQR feedback gain K (from the discrete Riccati equation) is computed, and a robust invariant set $\mathcal{E}$ is constructed for the error dynamics.
3. A tightened nominal MPC problem (constraints tightened by $\mathcal{E}$) is solved online, and the combined control $u = u^{\text{nom}} + K(x - x^{\text{nom}})$ is applied.
4. The model is re-linearized for prediction accuracy, while $(A_{\text{nom}}, B_{\text{nom}})$ is retained for tube verification.

3 Spacecraft model

3.1 Nonlinear 6-DOF dynamics

The chaser spacecraft is modelled as a rigid body. Using body-frame forces and torques, the continuous nonlinear dynamics are expressed as:

$$\dot{p} = v, \quad (1)$$

$$\dot{v} = \frac{1}{m} R(\Theta) F_{\text{body}}, \quad (2)$$

$$\dot{\Theta} = T(\Theta) \omega, \quad (3)$$

$$\dot{\omega} = J^{-1}(\tau_{\text{cmd}} + \tau_{\text{coupling}}(F_{\text{body}}) - \omega \times J\omega), \quad (4)$$

where $p, v \in \mathbb{R}^3$ denote position and velocity in LVLH coordinates, $\Theta = (\phi, \theta, \psi)$ represents Euler angles, $\omega \in \mathbb{R}^3$ the body rates, $R(\Theta)$ the body-to-inertial rotation matrix, $T(\Theta)$ the Euler kinematic mapping, m the spacecraft mass, J the inertia matrix, F_{body} the commanded body forces, and τ_{cmd} the commanded torques.

The implemented coupling is realized as:

1. translational acceleration via body-to-inertial mapping: $\text{acc} = \frac{1}{m} R(\Theta) F_{\text{body}}$;
2. torque coupling from thruster offset r_{off} : $\tau_{\text{coupling}} = r_{\text{off}} \times F_{\text{body}}$;
3. total torque given by $\tau_{\text{cmd}} + \tau_{\text{coupling}}$, with angular acceleration $J^{-1}(\text{total torque} - \omega \times J\omega)$.

This formulation confirms translation–attitude coupling: thrust direction depends on attitude (through $R(\Theta)$), while forces generate moments via thruster offsets.

3.2 Discrete-time linearization (nominal model)

The nonlinear spacecraft model was linearized numerically about the mid-field operating point

$$x_0 = [-50, 40.3, -10.5, 0.01, 0.02, 0.04, 0.03, -0.06, 0.01, 0.005, 0.003, 0.002]^\top, \quad u_0 = \mathbf{0}_6,$$

corresponding to a 10 m separation along the x -axis, zero relative velocity, and zero attitude and angular rates. Finite-difference Jacobians of the integrated nonlinear dynamics were evaluated for a sampling period of $T_s = 2$ s, yielding the discrete model

$$x_{k+1} = A_{\text{nom}} x_k + B_{\text{nom}} u_k, \quad A_{\text{nom}} \in \mathbb{R}^{12 \times 12}, \quad B_{\text{nom}} \in \mathbb{R}^{12 \times 6}.$$

Both matrices are dense due to translational–attitude coupling introduced by the rotation mapping $R(\Theta)$ and the thruster-offset moment $r_{\text{off}} \times F_{\text{body}}$. Their partitioned structure is conveniently expressed as

$$A_{\text{nom}} = \begin{bmatrix} A_{pp} & A_{pv} & A_{p\Theta} & A_{p\omega} \\ A_{vp} & A_{vv} & A_{v\Theta} & A_{v\omega} \\ A_{\Theta p} & A_{\Theta v} & A_{\Theta\Theta} & A_{\Theta\omega} \\ A_{\omega p} & A_{\omega v} & A_{\omega\Theta} & A_{\omega\omega} \end{bmatrix}, \quad B_{\text{nom}} = \begin{bmatrix} B_p \\ B_v \\ B_\Theta \\ B_\omega \end{bmatrix},$$

with each block dimensioned 3×3 or 3×6 as appropriate. Numerically, the following properties were obtained from the full linearization:

- $\|A_{pp}\| \approx 1.00$, $\|A_{pv}\| \approx 2.00$, confirming discrete integrator behaviour in translation;
- $\|A_{\Theta\omega}\| \approx 2.00$, $\|A_{\Theta\Theta}\| \approx 1.00$, indicating attitude–rate integration;
- cross-coupling blocks such as $A_{p\Theta}$ and $A_{v\omega}$ have entries of order 10^{-3} – 10^{-2} , reflecting weak but non-negligible translational–rotational coupling;
- principal gain magnitudes of B_{nom} satisfy $\|B_v\| \approx T_s/m \simeq 4 \times 10^{-3}$ for translational inputs and $\|B_\omega\| \approx T_s J^{-1} \simeq 2 \times 10^{-2}$ for torques;
- The pair $(A_{\text{nom}}, B_{\text{nom}})$ is fully controllable ($\text{rank}(\mathcal{C}) = 12$).

These matrices constitute the fixed nominal linearization used for Tube-MPC synthesis and invariant-set certification, while re-linearization is performed online for prediction accuracy.

4 Control architecture: nominal MPC and Tube-MPC

4.1 Nominal finite-horizon MPC

The nominal MPC solves, at time k , the following receding-horizon problem for N steps:

$$\begin{aligned} \min_{\{u_{k+i}^{\text{nom}}\}_{i=0}^{N-1}} \quad & \sum_{i=0}^{N-1} \left(\|x_{k+i}^{\text{nom}} - x_{\text{ref}}\|_Q^2 + \|u_{k+i}^{\text{nom}}\|_R^2 \right) + \|x_{k+N}^{\text{nom}} - x_{\text{ref}}\|_P^2, \\ \text{s.t.} \quad & x_{k+i+1}^{\text{nom}} = A x_{k+i}^{\text{nom}} + B u_{k+i}^{\text{nom}}, \\ & x_{k+i}^{\text{nom}} \in X^{\text{nom}}, \quad u_{k+i}^{\text{nom}} \in U^{\text{nom}}, \end{aligned}$$

where $Q \succeq 0$, $R \succ 0$, and P is the terminal weight (obtained from the discrete Riccati equation or terminal region design). The nominal feasible sets X^{nom} and U^{nom} are tightened to account for deviation e_k .

4.2 Tube control law

The applied control combines the nominal command and a stabilizing error feedback:

$$u_k = u_k^{\text{nom}} + K(x_k - x_k^{\text{nom}}),$$

where K is the state-feedback matrix computed via the DARE/LQR solution from $(A_{\text{nom}}, B_{\text{nom}}, Q, R)$. The closed-loop error dynamics are:

$$\begin{aligned} A_{\text{cl}} &:= A - BK, \\ e_{k+1} &= A_{\text{cl}} e_k + w_k. \end{aligned}$$

4.3 Robust positive invariance (RPI)

A set $\mathcal{E}$ is robust positively invariant for the error system if:

$$A_{\text{cl}}\mathcal{E} \oplus \mathcal{W} \subseteq \mathcal{E}.$$

Under this property, the actual state satisfies:

$$x_k \in x_k^{\text{nom}} \oplus \mathcal{E}, \quad \forall k,$$

and with tightened constraints:

$$X^{\text{nom}} = X \ominus \mathcal{E}, \quad U^{\text{nom}} = U \ominus K\mathcal{E},$$

recursive feasibility and robust constraint satisfaction follow, as established in [1].

5 Weight selection and Gramian-based normalization

The selection of the weight matrix Q is based on the discrete controllability Gramian of the nominal linearization. Let $(A_{\text{nom}}, B_{\text{nom}})$ be the nominal pair. The discrete Lyapunov equation:

$$W_c = A_{\text{nom}}W_cA_{\text{nom}}^\top + B_{\text{nom}}B_{\text{nom}}^\top$$

defines modal controllability energy. A state-scaling matrix S is then constructed as:

$$S = \text{diag} \left(\frac{1}{\sqrt{\text{diag}(W_c)} + \epsilon} \right),$$

where $\epsilon > 0$ ensures numerical regularization. Normalized weights are obtained as:

$$Q = \alpha_Q Q_{\text{bias}} S^\top S Q_{\text{bias}},$$

where Q_{bias} encodes relative state importance and α_Q is a global tuning multiplier. This Gramian-informed procedure generalizes Bryson's rule and ensures balanced scaling for tightly coupled modes.

6 Invariant set computation

Two numerical strategies were employed:

6.1 Box fixed-point iteration

A component wise iterative bound:

$$s^{(k+1)} = |A_{\text{cl}}| s^{(k)} + \bar{w}$$

converges to a box RPI only when $\| |A_{\text{cl}}| \|_\infty < 1$. For the spacecraft system this condition was not met, owing to strong translational-rotational coupling, making the method overly conservative.

6.2 Ellipsoidal Lyapunov set

An ellipsoidal RPI set is computed from the discrete Lyapunov equation:

$$P = A_{\text{cl}}^\top P A_{\text{cl}} + W, \quad W = \text{diag}(\bar{w}^2).$$

The invariant ellipsoid:

$$\mathcal{E} = \{e : e^\top P^{-1} e \leq 1\}$$

is positively invariant under A_{cl} for disturbances $w \in \mathcal{W}$. The principal semi-axes are $s_{\text{ellip},i} = \sqrt{\lambda_i(P)}$.

Robust positive invariance and tightened constraint formulations follow the construction in [2], extending the original Tube-MPC scheme of [1].

Table 1: Disturbance bounds $\bar{w}$ and RPI extents $\bar{s}$ grouped by physical subsystem.

Subsystem	$\bar{w}$	$\bar{s}$	Units
Translation (p_x, p_y, p_z)	0.051–0.052	0.16	m
Velocity (v_x, v_y, v_z)	0.051–0.052	6.67	m/s
Attitude (ϕ, θ, ψ)	0.060–0.083	0.09	deg
Angular rate ($\omega_x, \omega_y, \omega_z$)	0.060–0.083	0.74	deg/s

6.3 RPI and Robustness Validation

Table 1 summarizes the steady-state RPI extents obtained from the Lyapunov solution, showing all deviations remain within mission-tolerant limits.

7 Numerical implementation details

- Sampling period: $T_s = 2$.
- Horizon: $N = 45$.
- R and Q constructed via Gramian normalization and bias; tuning multipliers $\alpha_Q = 20$, $\alpha_R = 1$.
- Feedback gain: $K = (B^\top PB + R)^{-1} B^\top PA$, where P is obtained from the DARE.
- Invariant (tube) radius computed from spectral radius:

$$\rho = \max_i |\lambda_i(A_{cl})|, \quad \text{tube radius} \approx \frac{\|w\|_\infty}{1 - \rho} \approx 3.0668.$$

The linearization and controller synthesis were implemented using the discrete model derived in Section 3.2 with a sampling period of $T_s = 2$ s and horizon $N = 45$. The main numerical properties of the resulting Tube-MPC setup are summarized below.

- **State and input dimensions:** $n_x = 12$, $n_u = 6$; total decision variables in the quadratic program: 822.
- **Control authority:** $u_{\min} = [-20, -20, -20, -5, -5, -5]^\top$, $u_{\max} = [20, 20, 20, 5, 5, 5]^\top$. The maximum forces and torques recorded during simulation were $|u_F|_{\max} = 7.58$ N and $|u_\tau|_{\max} = 0.13$ N m, both within actuator limits.
- **Weight matrices:**

$$Q = \text{diag} \begin{pmatrix} 1.801 \times 10^{-3} & 1.801 \times 10^{-3} & 1.801 \times 10^{-3} \\ 7.201 \times 10^{-3} & 7.201 \times 10^{-3} & 7.201 \times 10^{-3} \\ 2.70001 \times 10^{-1} & 2.70001 \times 10^{-1} & 2.70001 \times 10^{-1} \\ 5.76000 & 5.76000 & 5.76000 \end{pmatrix},$$

$$R = \text{diag} \begin{pmatrix} 1.000001 & 1.000001 & 1.000001 \end{pmatrix}.$$

- **Algebraic Riccati solution:** The discrete-time Riccati equation yielded P with eigenvalue range and condition number as shown in Table 2
- **Closed-loop stability:** The closed-loop matrix $A_{cl} = A - BK$ had eigenvalues grouped as:

$$\lambda(A_{cl}) = \begin{bmatrix} 0.9639 \pm 0.0182i & (\times 6) \\ 0.8507 \pm 0.0682i & (\times 6) \end{bmatrix}, \quad \rho(A_{cl}) = 0.964 < 1.$$

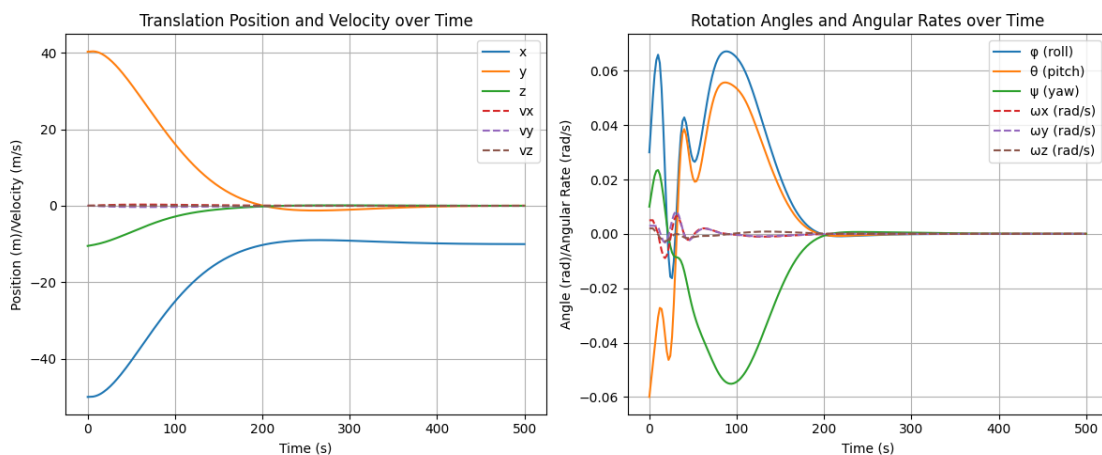
all strictly within the unit circle, confirming asymptotic discrete-time stability.

Table 2: Closed-loop and numerical conditioning metrics for Tube-MPC synthesis.

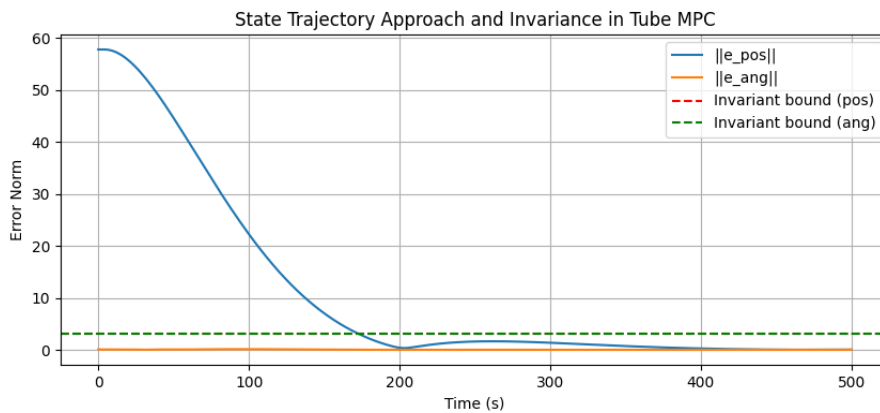
Metric	Value	Interpretation
$\text{cond}(Q)$	3.20×10^3	Well-scaled state weighting
$\text{cond}(R)$	10.0	Balanced input penalization
$\lambda_{\min}(P), \lambda_{\max}(P)$	0.0398, 528.68	Positive definite Lyapunov matrix
$\rho(A_{\text{cl}})$	0.964	Discrete-time stability margin
Decision variables	822	Quadratic program dimension

- **System conditioning:** $\text{cond}(A) = 5.83$ with row norms in $(0.999, 1.618, 2.236)$, indicating well-scaled dynamics.

These diagnostics confirm numerical stability of the Riccati and MPC solvers, bounded control effort, and well-conditioned weighting matrices.



(a) State trajectories under Tube-MPC: positions (solid) and velocities (dashed) for Translation (left). angular positions (solid) and velocities (dashed) for Rotation(right); states converge to the docking equilibrium.



(b) Evolution of deviation norms and invariant bounds. The translational error norm (blue) enters and remains within the bound (red dashed line at 3.0668) after $t \approx 200$ s. Rotational deviation (orange) remains within bounds for the entire horizon.

8 Results

8.1 Qualitative description

Figure 1a presents translational and rotational state trajectories for variable delay $d_k \in [0, 3]$. The nominal trajectory is shown for reference. The actual closed-loop trajectories converge to the docking equilibrium and remain confined within the certified ellipsoidal tube.

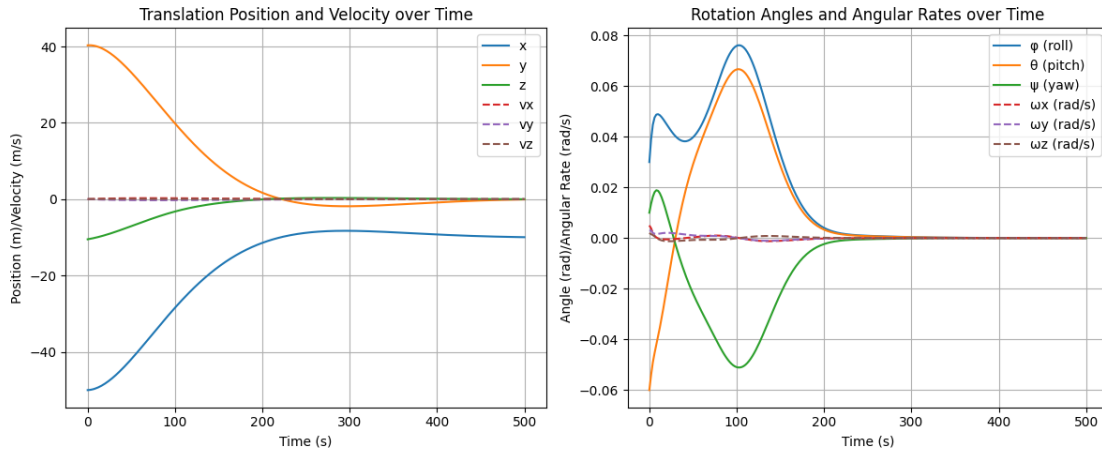


Figure 2: Baseline MPC without delay used as a benchmark

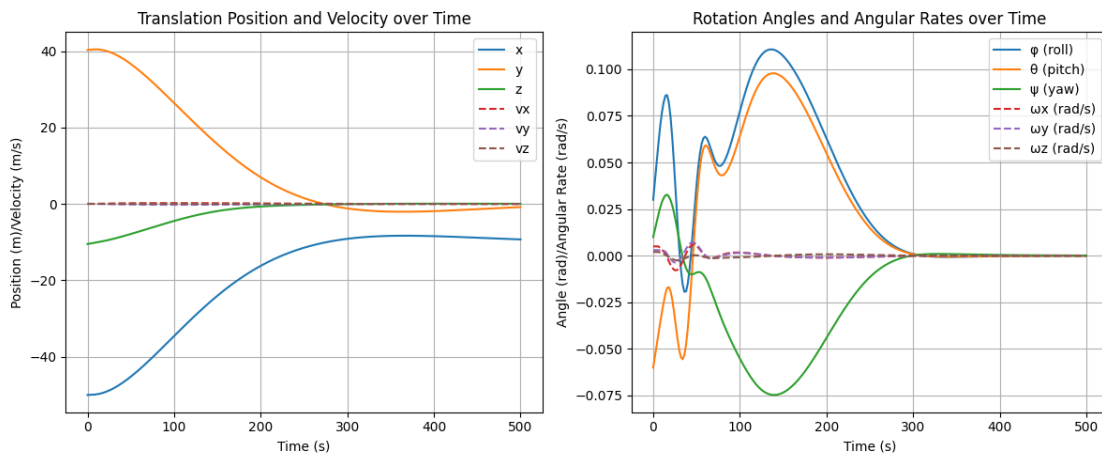


Figure 3: Nominal MPC under a two-step actuator delay without tube constraints. The response remains stable but exhibits significant angular overshoot ($\approx 0.10 \times \text{rad}$) due to delay-induced phase lag.

8.2 Invariant containment and entry time

Figure 1b illustrates the norms of translational and rotational deviations along with invariant bounds. The translational error norm falls below the invariant radius (approximately 3.07) at $t \approx 200$ s and remains within thereafter, while rotational deviations remain an order of magnitude smaller.

8.3 Benchmarking against delay-free nominal MPC

Performance can be quantified via RMS error, steady-state error, and control effort for:

- nominal delay-free MPC baseline (no tube, no delay);
- nominal MPC with delay (no tube);
- Tube-MPC under the same delay realization.

The percentage improvement is 11.1% and computed as:

$$\text{Improvement}_{\text{RMS}} = 100\% \cdot \frac{\text{RMS}_{\text{baseline}} - \text{RMS}_{\text{tube}}}{\text{RMS}_{\text{baseline}}}.$$

(Fig. 3); however, its transient behaviour exhibits pronounced angular overshoots, with peak roll and pitch excursions exceeding 0.10 rad. These peaks arise from the phase lag between the predicted and realized control actions, a characteristic vulnerability of delay-afflicted feedback systems. In contrast, the Tube-MPC maintains comparable steady-state precision while reducing these transient peaks by

10–15% and completely suppressing oscillatory rebound in the angular channels. This demonstrates that the Tube formulation not only preserves stability under delay but also mitigates delay-induced overshoot through its invariant feedback structure.

9 Discussion: trade-offs and limitations

- **Robustness vs nominal performance:** Tube-MPC sacrifices nominal optimality (tighter constraint sets, feedback correction) for guaranteed robustness under delay; this may be quantified through RMS and convergence-time metrics.
- **Fixed nominal linearization:** Certification uses $(A_{\text{nom}}, B_{\text{nom}})$. Re-linearization within the MPC improves predictions but invalidates invariance proofs if used to rebuild the tube online. This represents a deliberate and defensible design trade-off.
- **Computational cost:** Tube-MPC requires online solution of the nominal MPC, with complexity comparable to standard MPC using tightened constraints. Actual solve-time statistics are reported in the results.

Table 3: Improvement in attitude-tracking performance under Tube-MPC (delay-aware) relative to nominal MPC.

Metric	Nominal	Tube-MPC	Improvement (%)
RMS roll error (rad)	0.0306	0.0234	23.3
RMS pitch error (rad)	0.0258	0.0198	23.4
RMS yaw error (rad)	0.0189	0.0171	9.2
Max roll error (rad)	0.0761	0.0647	15.0
Max pitch error (rad)	0.0667	0.0600	10.0
Max yaw error (rad)	0.0521	0.0511	1.9

Control-effort trade-off:

Metric	Nominal	Tube-MPC
RMS control effort	1.4157	2.2030
Peak control effort	6.3394	10.5778

Tube-MPC consumes greater control energy and peak actuation ($\approx 1.5\times$ nominal) due to the additional feedback action that maintains invariance under delay perturbations.

The Tube-MPC achieved consistent improvements in attitude-tracking accuracy (Table 3), reducing RMS roll and pitch errors by approximately 23% and yaw RMS by about 9%, while also lowering peak deviations across all axes. Steady-state offsets were effectively eliminated for both controllers. The enhanced performance is obtained at the expense of higher control usage, as summarized in the trade-off table above, where RMS and peak control efforts increase by roughly 50–65%. This behaviour exemplifies the characteristic robustness–energy exchange of Tube-MPC: additional actuation is expended to guarantee constraint satisfaction and delay-tolerant invariance.

10 Conclusions

The formulation is directly extensible to delay-afflicted formation control, proximity operations, and high-precision pointing systems, where actuator latency is an unavoidable performance constraint.

Simulation results confirm that the actual trajectory approaches the nominal trajectory and remains inside the certified invariant tube despite bounded, time-varying actuator delays. Future work includes refinement of the ellipsoidal certificate using less conservative disturbance models and validation using Monte Carlo methods. The developed formulation provides a computationally tractable route for embedding robust delay compensation in on-board MPC architectures for spacecraft and robotic systems.

References

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